

PROJECT DOCUMENTATION

Sales Intelligence Dashboard

Executive BI asset synthesizing six years of transactional data across Power BI and Tableau, extended with county-level Kenyan market intelligence.

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ORGANISATION	skynet-datagrid-labs
PLATFORMS	Microsoft Power BI · Tableau Desktop
STACK	DAX · Power Query · LOD Expressions
DATA PERIOD	6 Years (Full Fiscal Cycles)
MARKETS COVERED	United States · United Kingdom · France · Kenya
REPOSITORY	github.com/skynet-datagrid-labs/sales-intelligence-dashbo ard

This edition of the documentation expands the original three-platform release with a full breakdown of every live dashboard, an updated data dictionary reflecting the newly added Kenyan county-level dataset, a development methodology narrative, governance notes, and a forward-looking roadmap.

skynet-datagrid-labs · Sales Intelligence Dashboard · 2024 – 2026

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1. Project Overview

This project is a production-grade business intelligence asset built to transform six years of fragmented transactional sales data into a unified executive intelligence layer. What began as a two-platform release (Microsoft Power BI and Tableau) has since grown into a six-dashboard portfolio spanning four distinct geographic markets – the United States, the United Kingdom, France, and, most recently, Kenya – while continuing to draw from a shared underlying analytical framework.

The Power BI implementation serves as an executive command center, offering multi-page reporting with regional filtering, category-level sales analysis, demographic segmentation, and year-over-year comparisons across 2019 to 2023. The Tableau implementation focuses on strategic analytics, leveraging advanced features including Level of Detail (LOD) expressions, sets, parameters, and map-based geographic analysis to surface insights that go beyond standard charting.

A third component, the Tableau KPI Performance Tracker, delivers state-level sales intelligence across the United States, featuring headline KPI cards, monthly trend lines, and an interactive choropleth map for geographic sales distribution. Two regional KPI dashboards follow the same design language for the United Kingdom and France, each pairing a summary metrics strip with a native choropleth map at county and department granularity respectively.

The newest addition, the Kenya Sales KPI Dashboard, extends this same framework to a 47-county soft drink sales dataset, giving the portfolio its first African market and demonstrating that the underlying design pattern – KPI strip, trend line, choropleth map, and ranked product/customer breakdowns – generalizes cleanly to a new geography with a different administrative boundary structure and a different product category. Together, the six dashboards demonstrate cross-platform BI engineering capability, multi-market analytical range, and redundancy for enterprise stakeholder preferences.

2. Strategic Objective

Raw transactional records, however voluminous, yield limited strategic insight in their unprocessed form. The core objective of this asset is to close the gap between data collection and actionable intelligence by providing three distinct capabilities to the business:

- **Executive Decision Support:** Real-time visibility into revenue health, regional performance, and customer segment dynamics, delivered through an interface designed

for non-technical stakeholders. KPI cards and conditional formatting allow an executive to assess business health in under ten seconds without needing to interpret a single formula.

- **Profitability Intelligence:** Multi-dimensional analysis across product categories, geographic territories, and temporal trends, enabling granular understanding of where margin is generated and lost. Region, company, and category filters let a user isolate a single dimension's contribution to profit without editing the underlying workbook.
- **Strategic Planning Foundation:** Historical pattern synthesis across six fiscal years, enabling evidence-based forecasting, seasonality modeling, and resource allocation decisions grounded in data rather than assumption.
- **Cross-Market Comparability:** A consistent visual and analytical grammar – the same KPI-strip, map, trend-line, and ranked-bar layout – applied identically across the US, UK, France, and Kenya, so a stakeholder who understands one regional dashboard can read any of the others without retraining.

3. Dataset Specification

Attribute	Detail
Temporal Coverage	6 years of full fiscal cycles
Data Source	Enterprise transactional database, supplemented with public boundary shapefiles
Record Volume	High-velocity sales transactions across four national markets
Update Cadence	Real-time synchronization (Power BI) / scheduled extract refresh (Tableau)
Geographic Markets	United States (state-level) · United Kingdom (county-level) · France (department-level) · Kenya (county-level, 47 counties)

The dataset is organized across five core analytical dimensions:

Dimension	Description	Analytical Application
Sales Transactions	Line-item revenue records	Revenue trend analysis, product performance
Customer Segments	Demographic and behavioral cohorts, segmented by central, east, south, and west territory files	Retention analysis, lifetime value modeling
Geographic Territories	Regional sales distribution across four markets, from state-level (US) down to county/department-level (UK, France, Kenya)	Market penetration, expansion opportunities
Product Categories	SKU-level classification, including a dedicated soft drink category used consistently across the UK, France, and Kenya dashboards	Portfolio optimization, cross-sell identification
Time Intelligence	Daily, weekly, monthly, quarterly data spanning 2017–2024 depending on market	Seasonality detection, year-over-year comparison

3.1 The Kenyan Dataset

The Kenya extension follows the same acquisition and modeling pattern established by the UK and France datasets: an accurate county-boundary shapefile set

(**kenya_counties_accurate.shp** and its companion **.cpg / .dbf / .prj / .shx** files) paired with a transactional CSV – **soft_drink_sales_kenya_47.csv** – covering soft drink sales across all 47 counties. A placeholder file is retained alongside the live extract to preserve the folder’s schema during development and refresh cycles, consistent with the convention used across the other regional folders.

Choosing soft drink sales as the anchor category for the Kenyan market keeps the new dashboard directly comparable to the existing US regional sales file (**soft_drink_sales.csv**), allowing a future cross-market comparison view to be built without reconciling mismatched product taxonomies.

4. Dashboard Architecture

The solution is structured across five engineering layers, each responsible for a distinct function in the data pipeline from source to visualization:

Layer	Components	Function
Data Ingestion	ETL pipelines, incremental refresh	Automated data extraction from source systems
Data Modeling	Star schema, relationship management	Optimized query performance, logical data organization
Calculation Engine	DAX measures, calculated columns; LOD expressions and table calculations in Tableau	Complex business logic, time intelligence functions
Visualization Layer	Interactive dashboards, drill-through pages, choropleth maps, ranked bar charts	Executive presentation, exploratory analytics
Security Layer	Row-level security (RLS), role-based access	Data governance, territory-based restrictions

5. Dashboard-by-Dashboard Breakdown

This section documents each of the six live demonstrations in the portfolio individually, covering platform, geographic scope, chart inventory, filter set, and the specific analytical techniques each one demonstrates.

5.1 Power BI – Executive Command Center

Platform: Microsoft Power BI · Markets: United States · Period: 2019–2023

A multi-page report built as the enterprise-distribution counterpart to the Tableau workbooks. Pages are organized around regional filtering, category-level sales analysis, and demographic segmentation, with year-over-year small multiples letting a viewer compare the same metric across five fiscal years without switching pages. DAX measures drive all time-intelligence calculations, and the report demonstrates the breadth of native Power BI visuals – matrix tables, small multiples, and slicer-driven cross-filtering – under a single consistent brand treatment.

5.2 Tableau – Strategic Intelligence

Platform: Tableau Desktop · Markets: Multi-region · Period: 2017–2018

A multi-sheet Tableau workbook built to demonstrate advanced analytical technique rather than broad market coverage. It combines region-wise profit distribution via stacked bar charts, a dual-axis sales-and-profit trend line, and a choropleth map for territory-level performance. The workbook is the primary showcase for Level of Detail (LOD) expressions, Tableau Sets, and parameter-driven analysis, and is intended to demonstrate cross-functional Tableau capability beyond standard charting.

5.3 Tableau – KPI Performance Tracker

Platform: Tableau Desktop · Markets: United States (state-level) · Period: Multi-year
Delivers state-level sales intelligence across the US with headline KPI cards for Sales, Profit, and Customer Count, a monthly sales trend line, a product-level bar chart with an average reference line, and an interactive choropleth map for geographic sales distribution. The average reference line is a deliberate touch: it lets a viewer see at a glance which products are over- or under-performing the portfolio mean, rather than just their raw rank.

5.4 Tableau – UK Sales KPI Dashboard

Platform: Tableau Desktop · Markets: United Kingdom (county-level) · Period: 2020 Q1 – 2023 Q4

The first of the three region-specific KPI dashboards. A dark KPI strip surfaces Revenue, Profit, Units Sold, and CoGS at a glance. Below it: a Profit-by-Month trend line, a Profit-by-County choropleth map, and ranked horizontal bars for Profit by Product and Profit by Customer. The left rail carries Region, Company, and Category multi-select filters plus a Quarter-of-Purchase range slider, and a radio-button metric toggle lets the viewer re-point every chart on the page at Revenue, Profit, or Units Sold without rebuilding a single view – a practical demonstration of parameter-driven dashboard design.

5.5 Tableau – France Sales KPI Dashboard

Platform: Tableau Desktop · Markets: France (department-level) · Period: 2020 Q1 – 2023 Q4

Structurally identical to the UK dashboard but operating at department-level geographic granularity across mainland France, giving it noticeably denser choropleth resolution. The KPI strip tracks Revenue, Profit, Units Sold, and CoGS; the supporting views are a CoGS-by-Month trend line, a CoGS-by-Department map, and auto-sorted ranked bars for CoGS by Product and CoGS by Customer. The metric toggle on this dashboard switches between Units Sold and CoGS, and the same Region, Company, Category, and Quarter filter set from the UK dashboard is retained for cross-market consistency.

5.6 Tableau – Kenya Sales KPI Dashboard

Platform: Tableau Desktop · Markets: Kenya (county-level, 47 counties) · Period: Aligned with the wider six-year window

The newest dashboard in the portfolio and the first to cover an African market. Built on the **soft_drink_sales_kenya_47.csv** extract and the accompanying county boundary shapefile, it follows the same design grammar established by the UK and France dashboards: a KPI strip, a county-level choropleth map of soft drink sales performance, a monthly trend line, and ranked product/customer breakdowns, with Region, Company, and Category filters carried over for consistency. Because Kenya's 47 counties are a finer and more numerous administrative unit than the UK's regions, this dashboard also serves as a stress test of the choropleth map's legibility at higher polygon density.

6. Key Performance Indicators

The asset tracks three tiers of KPIs, each serving a different organizational function:

6.1 Executive KPIs

Metric	Definition	Strategic Value
Total Revenue	Gross sales volume (current period)	Overall business health indicator
Revenue Growth	Year-over-year and quarter-over-quarter change	Trajectory assessment
Gross Margin	Revenue minus cost of goods sold	Profitability efficiency
Regional Profitability	Margin contribution by territory	Resource allocation optimization
Customer Lifetime Value	Projected future revenue per customer	Retention investment decisions

6.2 Operational KPIs

Metric	Definition	Tactical Application
Average Order Value	Revenue per transaction	Pricing strategy evaluation
Sales Cycle Duration	Lead-to-close timeline	Pipeline efficiency assessment
Win Rate	Closed-won over total opportunities	Competitive positioning analysis
Regional Penetration	Market share by territory	Expansion opportunity identification

6.3 Geographic & Cross-Market KPIs

Introduced alongside the UK, France, and Kenya dashboards, this tier supports comparison across markets rather than within a single one.

Metric	Definition	Strategic Value
Cost of Goods Sold (CoGS) by Territory	Total product cost attributable to a county, department, or state	Identifies where margin compression is occurring geographically

Metric	Definition	Strategic Value
Units Sold per County/Department	Transaction volume at the finest available administrative boundary	Local demand sensing, distribution planning
Cross-Market Category Share	A product category's share of total sales, compared across the US, UK, France, and Kenya soft-drink datasets	Portfolio strategy for categories sold in multiple markets

7. Technology Stack

Layer	Technology	Purpose
Platform 1	Microsoft Power BI	Executive dashboards, enterprise distribution
Platform 2	Tableau Desktop and Server	Strategic analytics, cross-functional collaboration
Calculation Language	DAX (Data Analysis Expressions)	Complex measures, time intelligence
Data Modeling	Power Query (M language)	ETL, data transformation
Geospatial Data	Shapefiles (.shp/.dbf/.prj/.shx/.cpg) for US, UK, France, and Kenya boundaries	Choropleth mapping at state, county, and department granularity
Security	Row-Level Security (RLS)	Access governance, territory restrictions
Version Control	GitHub	Asset versioning, collaboration tracking

8. Dashboard Capabilities

Capability	Implementation	Business Value
Real-Time Monitoring	Live data connections, automatic refresh	Immediate anomaly detection
Drill-Down Analysis	Hierarchical navigation (region, territory, account)	Root cause identification
What-If Parameters	Dynamic scenario modeling	Strategic planning simulation
Forecasting	Time series decomposition	Predictive revenue planning
Executive Summaries	KPI cards, trend indicators, conditional formatting	At-a-glance performance assessment
Mobile Optimization	Responsive layouts, phone report designer	On-the-go executive access

Capability	Implementation	Business Value
Metric Toggling	Radio-button parameters switching the driving measure across every chart on a page at once (UK, France, and Kenya dashboards)	One dashboard serves multiple analytical questions instead of three separate builds

9. Data Governance & Security

As the portfolio has grown to cover four national markets, governance has been kept deliberately simple and consistent rather than market-specific, to avoid the maintenance burden of four divergent security models.

- **Row-Level Security (RLS):** Applied at the territory level in the Power BI model so that a filtered view can, in principle, be scoped to a single region without exposing the full dataset.
- **Placeholder Files:** Each data folder (Customer, Product, Sales, UK, France, US, Kenya) retains a placeholder CSV that preserves the expected schema independently of the live extract, reducing the risk of a broken relationship when a source file is refreshed or temporarily removed.
- **Version Control:** All workbooks, the Power BI file, and every dataset are tracked in the **skynet-datagrid-labs** GitHub organisation, giving the project a full audit trail of when each regional dataset (most recently, Kenya) was introduced.
- **Attribution & Boundary Accuracy:** Geospatial files are labelled ***_accurate** (e.g. **kenya_counties_accurate.shp**, **france_mainland_accurate.shp**) to distinguish verified boundary sources from earlier draft shapefiles during development.

10. Development Methodology

Each regional KPI dashboard has been built to the same repeatable four-stage process, which is what allowed the Kenya dashboard to be added without redesigning the portfolio's visual language:

- **1. Source & Verify Boundaries:** Acquire an administrative boundary shapefile for the target market (county, department, or state), and validate it against the transactional dataset's own territory labels before building any visualization.
- **2. Model the Transactional Extract:** Land the sales CSV into the shared star schema, mapping it onto the existing Product, Customer, and Time dimensions so the new market inherits every existing calculated field automatically.
- **3. Build the Standard View Set:** Reproduce the fixed set of views – KPI strip, choropleth map, monthly trend line, and ranked Product/Customer bars – that every regional dashboard in the portfolio shares, rather than designing bespoke charts per market.
- **4. Wire Up Filters & Parameters:** Attach the shared Region/Company/Category filter set and the metric-toggle parameter last, once every view exists, so the toggle can be validated against all charts simultaneously.

11. Challenges & Lessons Learned

- **Choropleth Density at Fine Granularity:** Kenya's 47 counties render as noticeably smaller polygons than the UK's regional map at the same canvas size, which required re-checking label legibility and tooltip precision rather than assuming the existing map settings would transfer unchanged.
- **Schema Consistency Across Markets:** Keeping the Kenyan soft drink extract aligned to the same column structure as the US `soft_drink_sales.csv` file took deliberate effort, but pays off directly in Section 6.3's cross-market KPI tier, which depends on identical category naming across datasets.
- **Placeholder Discipline:** Maintaining a placeholder CSV in every data folder, including the new `kenya/` folder, proved useful during iterative development, since it kept relationships in the workbook intact while the live extract was still being finalized.

12. Future Roadmap

- **Cross-Market Comparison View:** A single Tableau dashboard overlaying soft drink performance across the US, UK, France, and Kenya on one normalized scale, using the KPI tier defined in Section 6.3.
- **Kenya Category Expansion:** Extending the Kenyan dataset beyond soft drinks into the same Alcoholic and Coffee categories already present in the UK dataset, to bring the two dashboards to full category parity.
- **Forecasting Layer:** Applying the time-series decomposition already used in the KPI Performance Tracker to the three regional KPI dashboards, adding a forward-looking projection alongside the existing historical trend lines.
- **Unified Data Dictionary:** Publishing the Section 13 data dictionary as a standalone, versioned document in the repository so it can be updated independently of this PDF whenever a new market or file is added.

13. Data Dictionary & Repository Structure

The table below documents every data folder currently maintained in the repository, including the newly added Kenyan dataset, for anyone extending the analysis or onboarding onto the project.

Folder	Key Files	Contents
Customer-Folder	Customer_central.csv, Customer_east.csv, Customer_south.csv, Customer_west.csv, customerplaceholder.csv	Customer records segmented by territory, used for demographic and behavioral cohort analysis
FranceSalesData	France_Sales_Mainland.csv, france_mainland_accurate.shp (+ .cpg/.dbf/.prj/.shx)	Mainland France transactional sales paired with a department-level boundary shapefile
Product-Folder	Product.csv, ProductData.xml, Productplaceholder.csv	SKU-level product classification shared across all regional dashboards
Sales-Folder	2023sales.csv, Salestill2022.csv, Salesplaceholder.csv	Core multi-year sales transaction history underlying the US and cross-market views
UK sales Data	UK_Sales_All_Counties_perfectly.csv, CTYUA_DEC_2024_UK_BUC.shp (+ .cpg/.dbf/.prj/.shp.xml), Placeholder.csv	UK county-level sales paired with the 2024 counties & unitary authorities boundary file
US regional sales	soft_drink_sales.csv, us_ak_hi_counties/states shift-conformal shapefiles	US soft drink sales with Alaska/Hawaii-adjusted state and county boundaries for accurate choropleth rendering
kenya	soft_drink_sales_kenya_47.csv , kenya_counties_accurate.shp (+ .cpg/.dbf/.prj/.shx), placeholder.csv	New: 47-county Kenyan soft drink sales paired with a verified county boundary shapefile
Root	mydata.csv, Business.twb, Sales_Intelligence_Dashboard.pdf , USSalesAnalysis2019-2024Version2.pbix	Combined working dataset, the primary Tableau workbook, this documentation, and the Power BI report file

The **kenya/** folder is the most recent addition and intentionally mirrors the file-naming convention already established by **UK sales Data** and **FranceSalesData**: a single accurately-labelled boundary shapefile set, one primary transactional CSV,

and one placeholder file. Maintaining this convention is what keeps onboarding a new market a matter of dropping in a folder rather than restructuring the model.

14. Repository & Access

The full project, including Power BI report files, Tableau workbooks, and all supporting assets, is maintained in the skynet-datagrid-labs GitHub organisation. The repository is version-controlled and open for collaboration.

Resource	Location
GitHub Repository	github.com/skynet-datagrid-labs/sales-intelligence-dashboard
Author	Tony Kenga (ORCID: 0009-0007-6899-8590)
Organisation	skynet-datagrid-labs

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